



Alberta Public Health Disease Management Guidelines

Cryptosporidiosis



This publication is issued under the Open Government Licence – Alberta (<http://open.alberta.ca/licence>). Please note that the terms of this licence do not apply to any third-party materials included in this publication.

This publication is available online at <https://open.alberta.ca/publications/cryptosporidiosis>

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without written permission of Alberta Health, Government of Alberta.

© Copyright of this document and its contents belongs to the Government of Alberta.

For further information on the use of this guideline contact:

Health.CD@gov.ab.ca

Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

Cryptosporidiosis | Alberta Health, Government of Alberta

© 2021 Government of Alberta | November 2021

Contents

Case Definition	4
Confirmed Case	4
Probable Case (Outbreak Only)	4
Reporting Requirements	5
Physicians, Health Practitioners and Others	5
Laboratories	5
Alberta Health Services and First Nations Inuit Health Branch	5
Epidemiology	6
Etiology	6
Clinical Presentation.....	6
Diagnosis	6
Treatment.....	6
Reservoir.....	6
Transmission	6
Incubation Period	7
Period of Communicability.....	7
Host Susceptibility	7
Incidence.....	7
Public Health Management.....	8
Key Investigation.....	8
Management of a Case.....	9
Management of Contacts	9
Preventive Measures.....	10
Appendix 1: Revision History	11
References	12

Case Definition

Confirmed Case

Laboratory confirmation of infection with or without clinical illness^(A) from an appropriate clinical specimen (e.g., stool, intestinal fluid or small bowel biopsy):

- Demonstration of *Cryptosporidium* oocysts

OR

- Detection of *Cryptosporidium* nucleic acid (e.g., PCR) in an appropriate clinical specimen

OR

- Detection of *Cryptosporidium* antigen by an approved method (e.g., EIA)

Probable Case (Outbreak Only)

Clinical illness^(A) in a person who is epidemiologically linked to a confirmed case.

^(A) Clinical illness is characterized by diarrhea (often profuse and watery) and abdominal cramps with or without loss of appetite, low grade fever, nausea, general malaise and vomiting. The illness may be prolonged in severely immunocompromised persons.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- The MOH (or designate) shall report all outbreaks to CMOH (or designate) via the [Alberta Outbreak Report Form \(AORF\)](#) as soon as possible using existing processes (e.g., CDOM or fax).
- For out-of-province and out-of-country cases, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Cryptosporidium parvum is an intracellular protozoan parasite. A ubiquitous pathogen, it is one of both medical and veterinary importance. *Cryptosporidium parvum* is the most prevalent species causing disease in humans.⁽¹⁾ Additional species names have been given when isolated from different hosts. It is known to infect and reproduce in the epithelial cell lining of the digestive or respiratory tracts of most vertebrates. *Cryptosporidium parvum* is a spore forming parasite whose lifecycle is completed within a single host.

Clinical Presentation

Asymptomatic infections with *C. parvum* are common and represent a source of infection for others. The major symptom is diarrhea, often profuse and watery, associated with abdominal cramping. Fever, malaise, anorexia, nausea, and vomiting occur but less often. The symptoms may come and go but, in general, abate within 30 days. In children, diarrhea may be preceded by anorexia and vomiting. Immunocompromised individuals (e.g., persons with AIDS) may not be able to clear the parasite. The disease can have a prolonged and fulminant course that may lead to death.

Diagnosis

Diagnosis is made through examination of stools, intestinal fluid or small bowel biopsy for oocysts or parasitic antigens. Detection of this organism is difficult unless it is looked for specifically. It may require more than one specimen, as shedding of oocysts is intermittent.

Treatment

- In most cases, cryptosporidiosis is a self-limited disease.
- There is no treatment of known value. For chronic cases, consultation with an infectious diseases physician (who may consider experimental or unproven therapies) is recommended.
- Rehydration and electrolyte replacement if indicated.

Reservoir

Humans, cattle, and sheep are the principal reservoirs for *Cryptosporidium*. Other domestic animals – including birds and reptiles, and occasionally wild animals – are also reservoirs for this parasite. Fertilizing salad vegetables with manure is known to be a source of human infection.

Transmission

Transmission of *Cryptosporidium* is via the fecal-oral route including person-to-person, animal-to-person or environmental, especially waterborne and foodborne. Transmission between sexual partners has been reported.

The parasite infects the intestinal epithelial cells. The end result is oocysts in feces that are able to survive under adverse environmental conditions for a long period of time. The oocysts are highly resistant to chemical disinfectants used to purify drinking water. The mode of transmission of *Cryptosporidium* increases the risk of infection for those living in group settings, travellers to endemic areas, and immunocompromised persons. Fewer than 10 organisms can cause disease.⁽²⁾

Incubation Period

The incubation period is not precisely known. It has been suggested that one to 12 days is the likely range, with an average of seven days.

Period of Communicability

Oocysts, the infectious stage of the parasite, appear at the onset of the symptoms and are infectious immediately upon excretion. The oocysts may be excreted in stool for weeks beyond the symptoms of the disease; however, in most cases the shedding of *C. parvum* stops within two weeks. These oocysts may remain infective outside the body for two to six months in a moist environment.

Host Susceptibility

Universal susceptibility. Persons with intact immune functions usually have asymptomatic or self-limited infections. It has been estimated that 10–20% of AIDS patients develop *C. parvum* infection at some time during their illness, and the *C. parvum* infection persists throughout.

Incidence

General

Cryptosporidium is considered one of the most common enteric pathogens in humans and domestic animals worldwide. Although it affects all age groups, children under six years of age, animal handlers, travellers, men who have sex with men, and close personal contacts of infected individuals (families, health care and daycare workers) are most likely to become infected.

In developing countries the prevalence ranges from 3–20%. The incidence is greatest in the summer and early fall (outdoor swimming season). Outbreaks in North America and Europe have been associated with contaminated drinking water; bathing in contaminated swimming pools, waterparks and lakes; and drinking unpasteurized apple cider contaminated with cow manure.^(1,2) The most notable outbreak occurred in 1993 in Wisconsin, when more than 400,000 people were infected via a contaminated water supply.⁽³⁾

Canada

Reference 3 applies to this section.

In 2000, 613 cases (rate 2.62/100,000) were reported. The highest incidence was in children one to four years of age, followed by children aged five to nine years. Recent outbreaks and cases in Canada have been associated with swimming in and drinking contaminated pool water.

Alberta

Reference 4 applies to this section.

From 1997 to 2000, a total of 289 cases were reported. In 2001, 443 cases were reported in the province. This was a significant increase when compared to previous years. The increased incidence in 2001 was associated with common source outbreaks occurring in a neighbouring province (141 cases), as well as contaminated local swimming pools.⁽⁵⁾ In 2002, 136 cases were reported. The number of cases continued to fall in 2003 and 2004, with 111 and 104 cases reported respectively.

Young children between one and 14 years of age had the highest incidence of disease (1997 to 2001).

Refer to the [Interactive Health Data Application \(IHDA\)](#) for more information.

Public Health Management

Key Investigation

- Confirm that the case meets the case definition.
- Obtain a history of illness including the date of onset, and signs and symptoms.
- Identify any underlying medical conditions that may increase host susceptibility.
- Determine the occupation of the case and if in a **sensitive situation or occupation (SSO)** that may pose a higher risk of transmission to others. See Table 1 for a definition of SSO.
- Determine the possible source of infection taking into consideration the incubation period^(B), reservoir, and mode of transmission. Assessment may include determining, identifying or obtaining:
 - a detailed food history including recent consumption of potentially contaminated food or water and the time of consumption,
 - attendance at daycare or institutions,
 - consumption of contaminated food or water, or other drink including unpasteurized milk;
 - contact with cattle, sheep, or other domestic animals,
 - recent visits to farms or petting zoos,
 - recent exposure to recreational water (treated or untreated),
 - recent travel or immigration,
 - residing in areas with poor sanitation including improper water treatment and sewage disposal, either in Canada or abroad, and/or
 - high risk sexual practices, especially contact with feces.
- Suspected contaminated food may be held and/or destroyed to prevent consumption.
- Assess for history of similar symptoms in other members of the household.
- Identify contacts, especially those in SSO that may pose a risk of transmission to others. (Refer to Table 1). Contacts include:
 - persons living in the household,
 - children and child care workers in a day care/day home, and
 - individuals exposed to the same source (if identified).

Table 1: Sensitive Situations or Occupations (SSO)

SSO	Activities
Food handler	• Touches unwrapped food to be consumed, <u>and/or</u> • Handles equipment or utensils that touch unwrapped food to be consumed.*
Health care, child care or other staff	• Has contact through serving food to <u>highly susceptible</u> patients or persons in whom an intestinal infection would have particularly serious consequences. • Provides direct patient care and is involved in the care of young children, elderly or dependent persons.
Child attending a child care facility or similar facilities	• Is diapered or unable to implement good standards of personal hygiene.
Any individual (older child or adult)	• Is unable to implement good standards of personal hygiene (e.g., with disabilities/challenges that may impact ability to perform good hand hygiene and is involved in an activity that may promote disease transmission.

* NOTE: Generally, food handlers who do not touch food, equipment or utensils in this way are not considered to pose a transmission risk; however, circumstances for each case should be assessed on an individual basis.

^(B) The incubation period of cryptosporidiosis is between 1–12 days. However, individual cases can vary, thus requiring flexibility when determining where and how the disease was likely acquired.

Management of a Case

- All cases should be advised of the following:
 - how the disease is transmitted, appropriate personal hygiene, routine infection prevention and control practices, and contact precautions,
 - to avoid food preparation until symptoms have resolved,
 - to practice safer sex and avoid sexual practices that facilitate fecal-oral transmission,
 - to avoid public recreational water (e.g., swimming pools) until after treatment is completed and diarrhea has resolved,
 - when possible, reduce or stop taking any immunosuppressive therapy, under the guidance of an infectious diseases physician.
- Contact precautions should be used in health care settings where children or adults have poor hygiene or incontinence that cannot be contained. Otherwise, routine practices are adequate.
- Refer to Table 2 for case exclusion criteria. For a summary of all enteric exclusions refer to [Enteric Transmission Risk Assessment and Exclusion Table](#).

Table 2: Case Exclusion

Cases	Category	Exclusion Criteria
Symptomatic	SSO	• The MOH may by order exclude the case until 48 hours after stools have returned to normal or the MOH is satisfied that the case is no longer infectious. – The case must be symptom free for 48 hours after stopping any antidiarrheal medication (if taken). – Lifting of exclusions is not conditional upon submission of stool specimens to demonstrate clearance of the organism. – Specimens may still be submitted on a case-by-case basis in consultation with the MOH. • If possible, consideration may be given to temporary redeployment away from activities that involve increased risk of transmission.
Asymptomatic	SSO	• Generally not required unless otherwise recommended by the MOH.
Symptomatic	Non-SSO	• No exclusion required; however, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped.
Asymptomatic	Non-SSO	• No exclusion required.

Management of Contacts

- Contacts should be instructed in disease transmission, appropriate personal hygiene, routine practices, and contact precautions.
- Symptomatic contacts should be assessed by a physician.
 - Contacts with positive stool specimens should be managed and treated as cases.
- Refer to Table 3 for contact exclusion criteria.

Table 3: Contact Exclusion

Contacts	Category	Exclusion Criteria
Symptomatic	SSO	• Same as a case. • Ensure the contact is assessed by their physician.
Symptomatic	Non-SSO	• No exclusion required however, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped. • Refer to their physician for assessment and testing, if required.
Asymptomatic	All	• No exclusion – contacts should monitor themselves for gastrointestinal symptoms, maintain good hand hygiene and food handling practices and seek medical attention if symptoms develop.

Preventive Measures

- Educate the **public** about the following:
 - personal hygiene, especially the sanitary disposal of items containing feces,
 - careful hand washing after defecation and sexual contact, and before preparing or eating food,
 - washing cutting boards, counter tops and utensils with soap and water after contact with raw meat
 - cooking meats thoroughly,
 - washing hands after contact with animals, animal feces, and animal environments, especially where the animals are ill with diarrhea,
 - methods of personal protective measures, in particular the correct and consistent use of condoms and discuss risk of sexual practices that permit fecal-oral contact,
 - accessing and drinking safe water supplies, and
 - testing private water supplies for presence of bacterial/parasitic contamination, if suspected.
- Educate **food handlers** about:
 - proper food and equipment handling, preparation, and hygiene, especially in avoiding cross-contamination from raw meat products, and
 - thorough hand washing before and after meal preparation.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology • Key Investigation section moved to Public Health Management (formerly called Control) • Added links where applicable
	Case Definition	• Added "(Outbreak Only)" to Probable Case definition as it is only notifiable in an outbreak
	Reporting	• Added a bullet on outbreak reporting
	Management of Case and Contacts	• Specified where terminology is SSO and included information in tables • Updated Exclusions to align with decisions agreed to in Enteric Transmission Risk Assessment and Exclusion Table • Re-worded to align with other enteric diseases
	Preventive Measures	• Reworded to align with other enteric diseases

References

1. Centres for Disease Control and Prevention. National Centre for Infectious Diseases, Division of Parasitic Diseases. *Cryptosporidiosis*. Parasites and Health. 2002.
2. US Food and Drug Administration Centre for Food Safety and Applied Nutrition. *Foodborne Pathogenic Microorganisms and Natural Toxins*. Bad Bug Book. 1992.
3. Public Health Agency of Canada. Notifiable Diseases On-Line – Cryptosporidiosis. August 2005.
4. Alberta Health and Wellness, Disease Control and Prevention. *Communicable Disease Reporting System*. March 2003.
5. Public Health Agency of Canada. *Waterborne Cryptosporidiosis Outbreak, North Battleford, Saskatchewan, Spring 2001*. Ottawa: CCDR 2001;27-22.
6. Public Health Agency of Canada. *An Outbreak of Cryptosporidium Parvum in a Surrey Pool with Detection in Pool Water Sampling*. Ottawa: CCDR 2004;30-07.